

Kevin Zhang

kevinkw.zhang@mail.utoronto.ca • +1-647-896-6853

LinkedIn: [kevin-zhang-uoft](#) • GitHub: [Diebrate](#)

Research interests

Optimal transport, stochastic differential equations, machine learning, scRNA-seq

Education

- 2019 – Present **University of Toronto** – Toronto, Ontario, Canada
PhD in Statistics
Advisors: [Dr. Dehan Kong](#), [Dr. Zhaolei Zhang](#). *GPA: 3.8.*
- 2015 – 2019 **University of Toronto** – Toronto, Ontario, Canada
HBSc in pathobiology, statistics and mathematics. *GPA: 3.89.*

Research experience

- 2023 – Present **Generative Modelling Using Zero-Shot Learning and Transfer Learning (on-going project)**
Supervisors: [Dr. Dehan Kong \(University of Toronto\)](#), [Dr. Zhaolei Zhang \(University of Toronto\)](#).
Currently designing novel numerical techniques that perform transfer learning among multiple batches of data (possibly embedded in different manifolds)
Making connections between transfer learning and classical statistical models (e.g., regression, classification).
- 2020 – Present **Trajectory Inference Using Stochastic Differential Equations and Mean Field Control (manuscript submitted, [preprint available](#))**
Supervisors: [Dr. Dehan Kong \(University of Toronto\)](#), [Dr. Zhaolei Zhang \(University of Toronto\)](#).
Proposed a comprehensive mathematical model integrating concepts from mean field control and SDE to capture crowd motion in a population observed over a timeline.
Constructed a neural network using PyTorch to parametrize the SDE model.
Implemented an innovative numerical algorithm more suitable for models with higher complexity as compared to other existing SDE solvers.
Analyzed multiple different data sets and demonstrated excellent customizability and performance from the model.

- 2021 – 2023 **Modelling Single-Cell Developmental Trajectory Using Optimal Transport (manuscript in preparation)**
 Supervisors: [Dr. Dehan Kong \(University of Toronto\)](#), [Dr. Zhaolei Zhang \(University of Toronto\)](#).
 Proposed a novel test statistic for change point detection in time series data and constructed an innovative mathematical model for population movement/
 Developed an integrative framework written in Python, R and MATLAB for numerical optimization and performance evaluation/
 The proposed testing method outperformed other state-of-the-art methods in change point detection.
 Analyzed high-dimensional ($\approx 20,000$ dimensions) time series data from over 250,000 cells.
- September 2017 **Mount Sinai Hospital - Kandel Lab (Project Student)**
 – April 2018 Supervisor: [Dr. Rita Kandel \(University of Toronto\)](#).
 Analyzed effects of Tgfb3 on the regeneration of nucleus pulposus tissue (cell/tissue culture and extracellular matrix quantification).
- May 2017 – **MaRS Discovery District - Mekhail Lab (Summer Student)**
 August 2017 Supervisor: [Dr. Karim Mekhail \(University of Toronto\)](#).
 Analyzed **effects of Poly-Q expansion on yeast replicative lifespan**(RNA variants and cell culture).

Publications

- 2018 **Conserved pbp1/ataxin-2 regulates retrotransposon activity and connects polyglutamine expansion-driven protein aggregation to lifespan-controlling rdna repeats.**
 L. Ostrowski, A. Hall, K. Szafranski, R. Oshidari, K. Abraham, J. Chan, C. Krustev, K. Zhang, A. Wang, Y. Liu, R. Guo, and K. Mekhail.
Communications Biology.

Teaching experience

Fall 2019-Present	Teaching assistant (University of Toronto) Courses include: STA130: An Introduction to Statistical Reasoning and Data Science STA220: The Practice of Statistics I STA255: Statistical Theory STA261: Probability and Statistics II STA302: Methods of Data Analysis I STA303: Methods of Data Analysis II STA304: Surveys, Sampling and Observational Data STA305: Design and Analysis of Experiments STA437: Methods for Multivariate Data STA442: Methods of Applied Statistics Held weekly tutorials/office hours and monitored online course forums. Invigilated and marked midterms and exams.
----------------------	---

Review experience

2023	Journal of the American Statistical Association (1 time)
2022	Journal of the Royal Statistical Society: Series C (1 time)

Talks and conferences

December 2022	Modelling Cellular Development Trajectory using Unbalanced Optimal Transport <i>Medicine by Design's 7th Annual Symposium</i>
July 2022	Modelling Single-Cell Type Trajectory Using Optimal Transport <i>The Fifth ICSA-Canada Chapter Symposium</i>

Honors and scholarships

2019-Present	Graduate Program Funding Package (Fully Funded PhD Program) (University of Toronto)
2018	New College Council In-Course Scholarship (University of Toronto)
2017	LMP Undergraduate Summer Research Program UROP Award (University of Toronto)
2015	Undergraduate Entrance Award (University of Toronto)

Technical skills

Programming languages

Proficient in: Python (e.g., numpy, pandas, pytorch), R, MySQL

Familiar with: C++, basic UNIX commands

Software

L^AT_EX, Git, Microsoft Office, Slurm Workload Manager

Languages

English (fluent), Chinese (native)